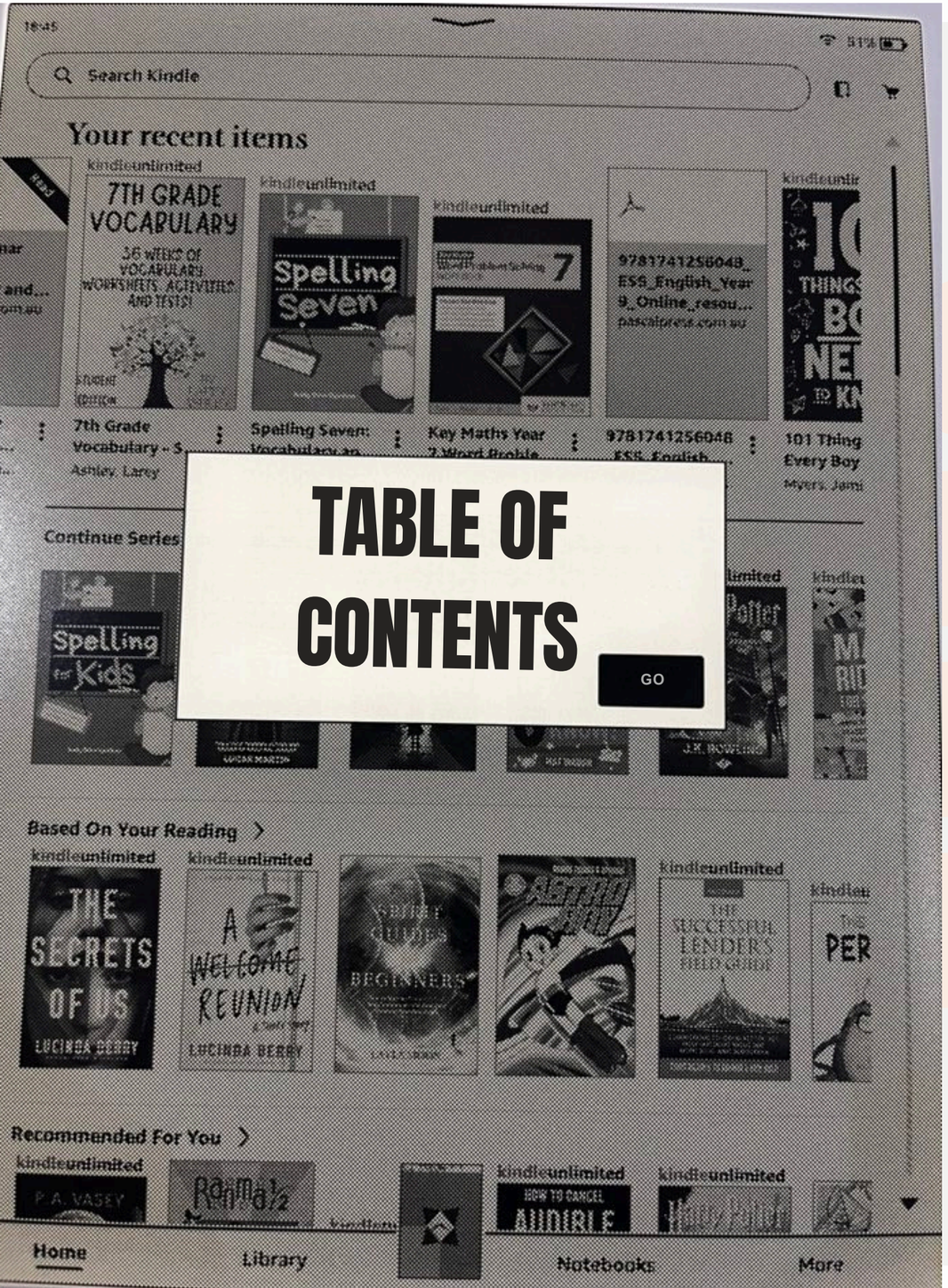
WESTERN SYDNEY
UNIVERSITY

PSO
Problem Solving in Organization



Perform Data Analysis for amazon kindle

PRESENT BY GROUP 2



01 Introduction

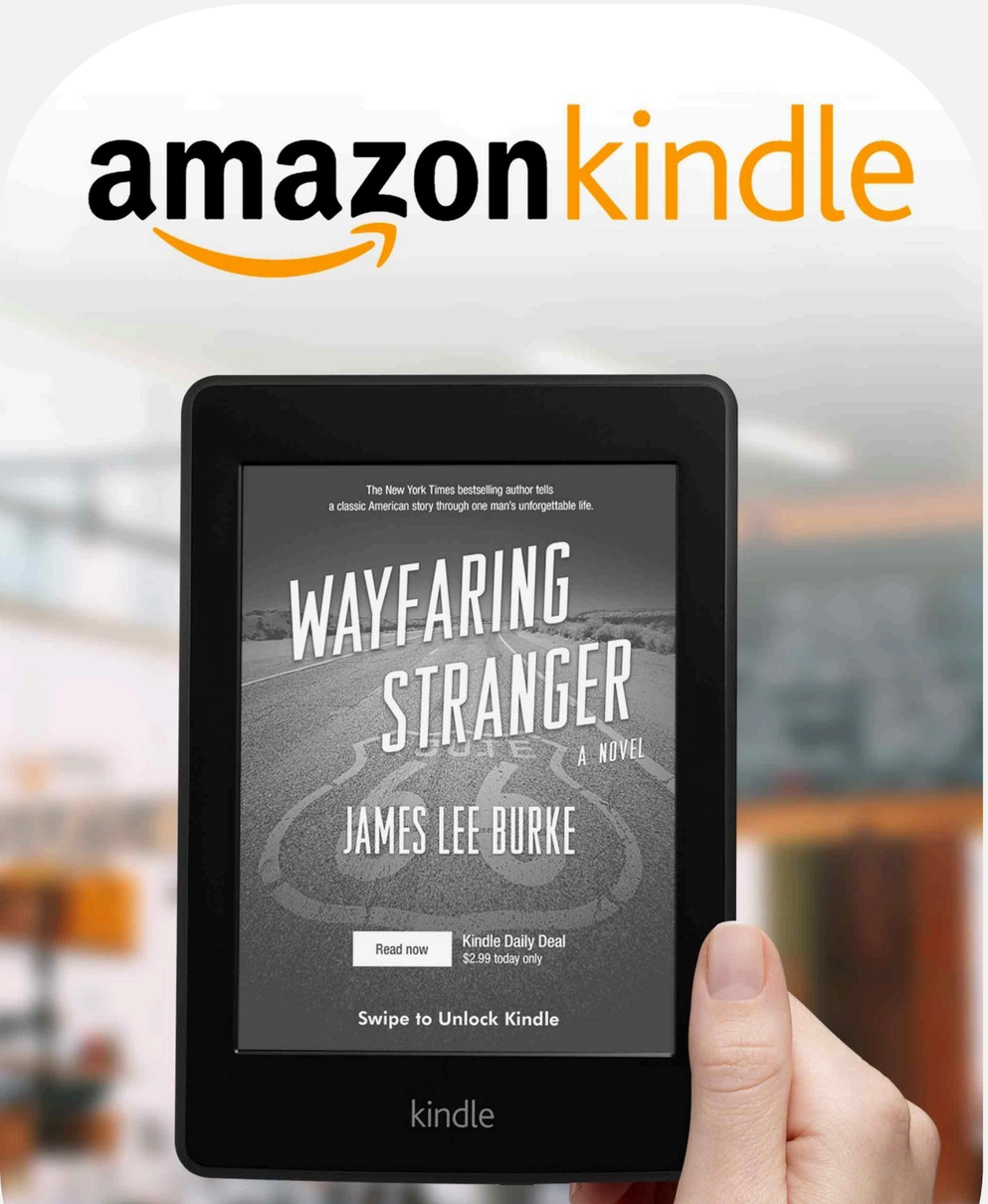
02 Data Description

03 Data Analysis

04 Proposed Solutions

INTRODUCTION

About Kindle



- **Kindle is Amazon's e-reader ecosystem launched in 2007, designed to make digital reading convenient, portable, and affordable.**
- **Kindle Unlimited – access to over 5 million titles with a monthly fee.**

Objective

- Explore and understand the Kindle dataset in order to uncover key patterns, trends, and relationships
- Generate actionable insights that can support business decisions in later stages

Approach

- Using the customer feedback as the foundation, allowed us to generate relevant questions and conduct meaningful analysis that aligns with the above objectives

DATA DESCRIPTION

Data Cleaning

- The data set contains **133,103 rows** and 16 columns showing the information of e-books on Kindle
- **No duplicates** is founded on data sets
- Fixed some errors are identified, such as Word font in “author”, “sold_by”
- **published_date**: have 7158 (36%) missing data -> Remove this column in analysis
- **author/sold-by:** have respectively 0.3% and 7% missing value -> Fill-in by **”NA**” value.
- “**price_range”, price_log(10)** is added to the data set to support the analysis.

No.	Column	Meaning	Qualitative Datatype	Quantitaive Datatype	Unique Value	Missing Value
1	asin	ID of each book	Norminal		133102	0
2	title	Name of book	Norminal		131834	0
3	author	Book's author	Norminal		72601	0
4	sold_by	Publishing company	Norminal		47	0
5	stars	User rating		Discrete	38	0
6	reviews	Customer reviews about the book		Discrete	8441	0
7	price	Cost of the book		Continuous	4795	0
8	is_kindle_unlimited	Is the book in Kindle Unlimited program	Binary		2	0
9	category_id	ID of book category	Ordinal		31	0
10	is_best_seller	Is it a best seller?	Binary		2	0
11	is_editors_pick	Selected as an Editor’s Pick?	Binary		2	0
12	is_good_reads_choice	Winner of Goodreads Choice Award?	Binary		2	0
13	category_name	Name of the book category	Norminal		31	0
14	price_range	Range of price - Group by Quantile 0-4.99($) 5-9.99($) 10-14.99 ($) 15-30.39($) 30.4 - 682($)	Ordinal		5	0
15	price_log(10)	Log 10 of price		Continuous	4794	0

DATA DESCRIPTION

Univariate

The dataset contains many outliers and skewed distributions, log transformation is needed for more accurate analysis

	Stars	Reviews	Price	Price_log(10)
Mean	4,4	887,38	15,13	0,98
Standard Error	0	13,99	0,06	0
Median	4,5	4	9,99	1
Mode	4,6	0	9,99	1
Standard Deviation	0,75	5104,88	22,25	0,4
Sample Variance	0,56	26059787,33	495,28	0,16
Kurtosis	26,54	2275,33	81,95	1,05
Skewness	-5	32,52	6,74	0,07
Range	5	618227,00	682,00	3,13
Minimum	0	0	0,00	-0,30
Maximum	5	618227	682	2,83
Sum	586193,2	118111491	2014382,61	130271,16
Count	133102	133102	133102	133102

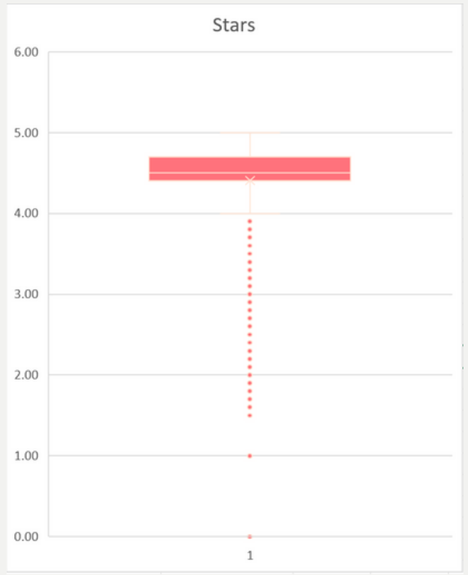
- **Price and reviews** are highly skewed with extreme outliers (some books have >600k reviews, prices up to 682 USD).
- **Log transformation** compresses large values, reduces skewness, and makes distributions closer to normal.

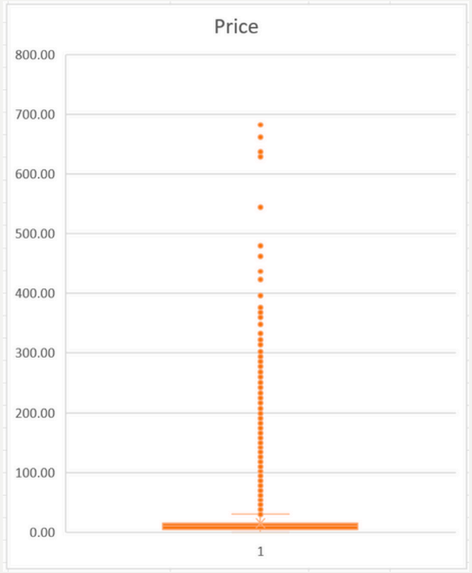
Outlier Threshold Calculation with log(10) - IQR Method			
	Star	**Reviews**	**Price**
Q1	4,4	0	0,7
Q2	4,5	4	1
Q3	4,7	365	1,18
IQR	0,3	365	0,48
Lower	3,95	-547,5	-0,02
Upper	5,15	912,5	1,89
Outlier	**0%**	**0%**	**0%**

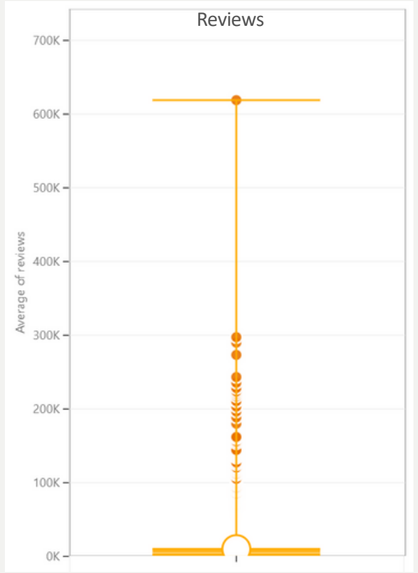
DATA DESCRIPTION

Correlation LOW

Star ratings show almost no correlation with either price or the number of reviews









	Stars	Reviews	Price	Price_log(10)
Stars	1			
Reviews	28.203.631	1		
Price	-126.704.778	-5.022	1	
Price_log(10)	-5.856.561	-5.935	735.069	1

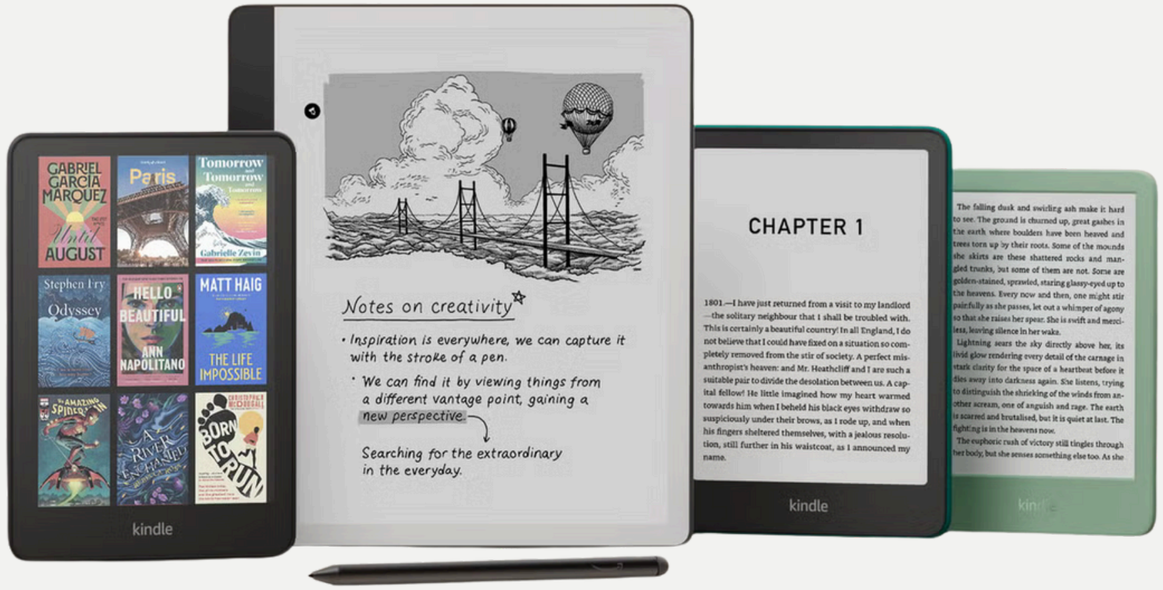
- **Correlation matrix** show that star ratings are largely independent of both price and the number of reviews.
- The number of reviews has only a **weak positive correlation** with price, and log(price) is strongly correlated with price.

DATA ANALYSIS

EDA Questions

How is Kindle Unlimited (KU) performing?

- **What are the top-performing categories for KU vs total books?**
- **What is the price range distribution for KU vs total books?**



User Engagement

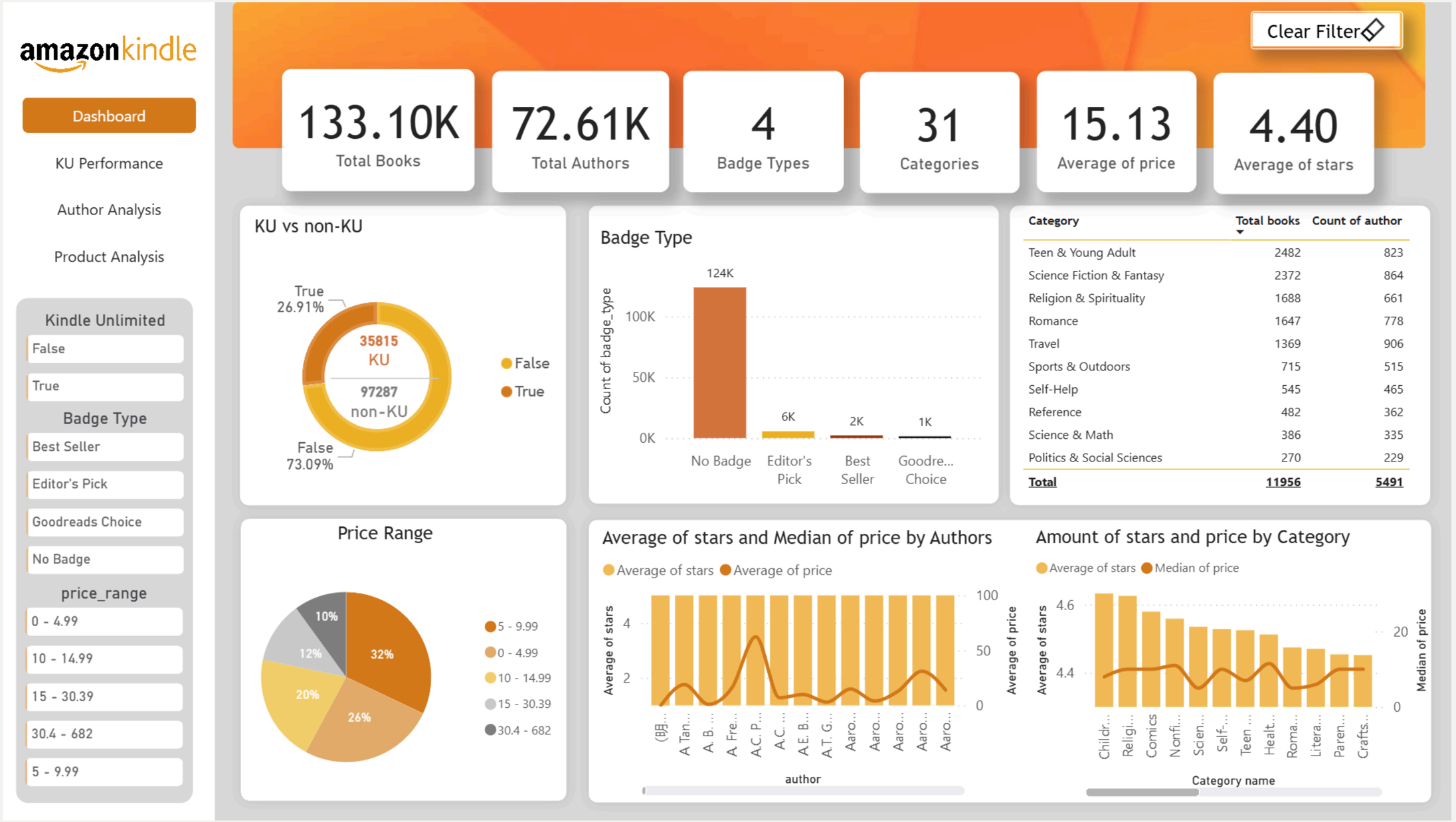
- Which authors dominate KU vs non-KU? Are KU authors more concentrated (few big names) or more distributed?
- Which group drives more reviews and engagement?

Product Performance

- Which books are the top performers (by reviews, ratings, or stars) within KU and outside KU?
- Do KU books rely more heavily on certain badges than non-KU books types?
- Are KU books more likely to be in the low-price / mid-price segments (0–9.99, 10–14.99) compared to high price?

DATA ANALYSIS

Overall Performance



DATA ANALYSIS

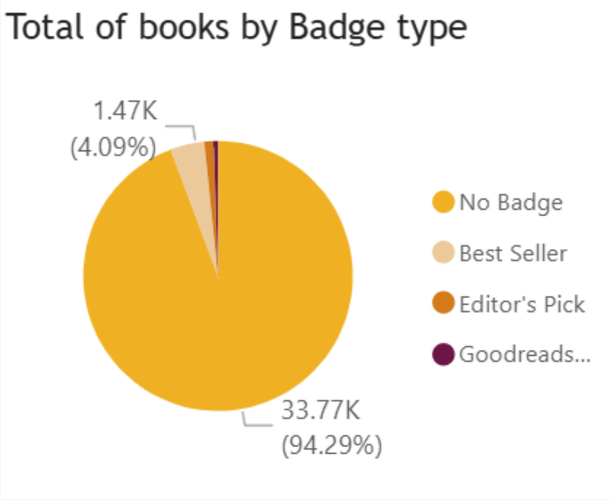
Kindle Unlimited

How is Kindle Unlimited (KU) performing?

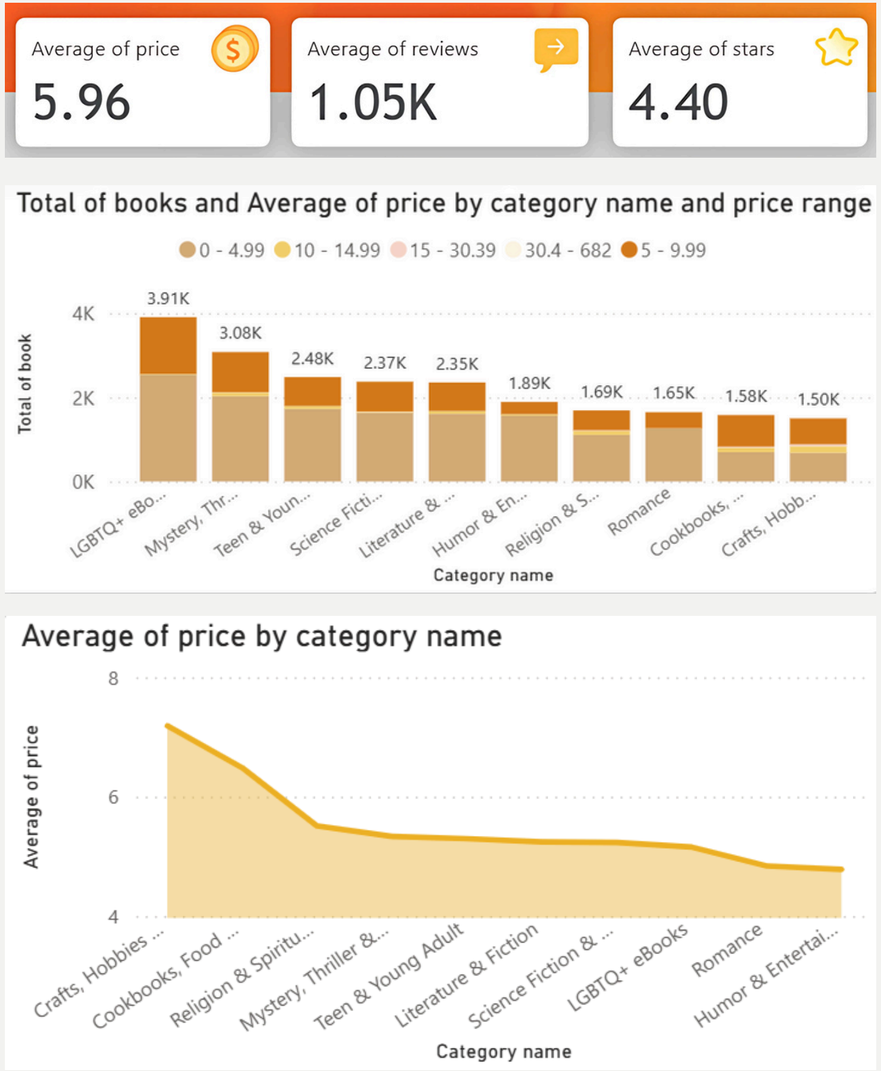
Current strategy: focus on famous **best seller** and **low price** books

Target: Increase no. of **user join KU**, KU customer loyalty, and KU engagement

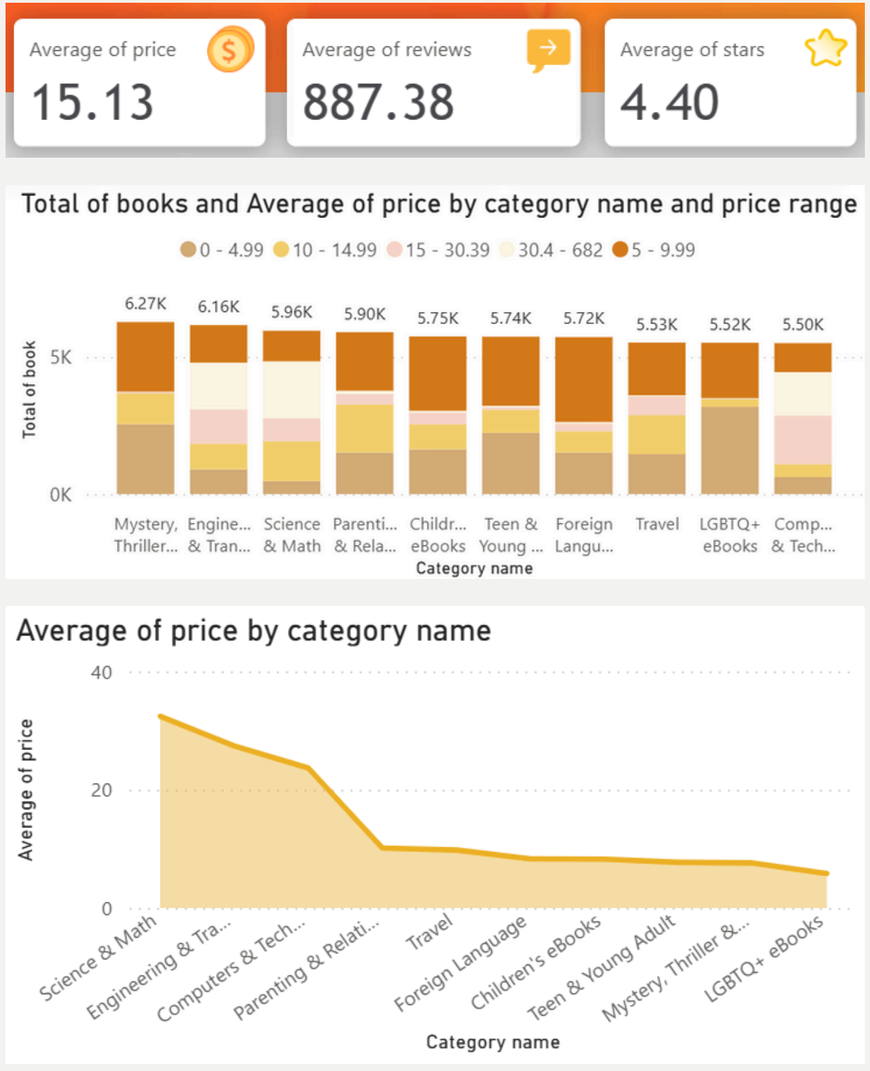
KU **replicate** badge distribution of total



KU BADGES DISTRIBUTION



KINDLE UNLIMITED



TOTAL

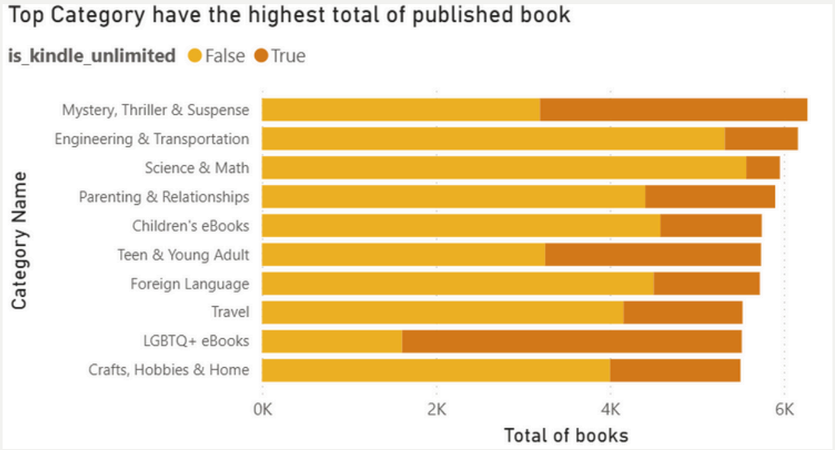
DATA ANALYSIS

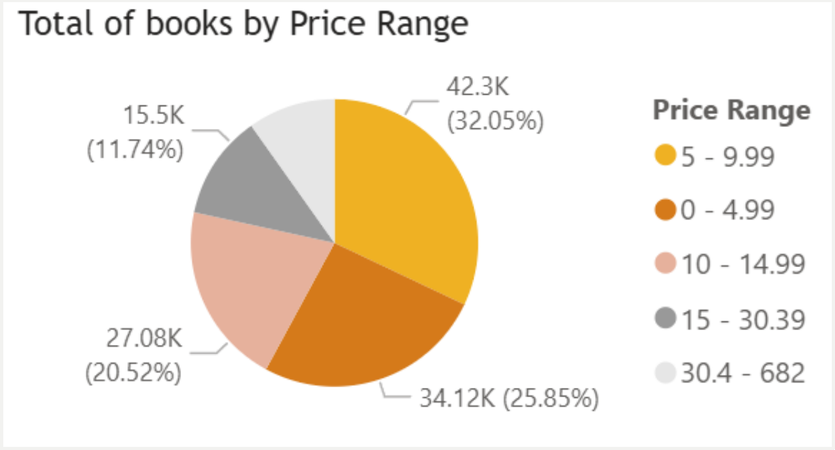
Product

- **Top performers (by reviews, ratings, or stars)** within KU and outside KU?
- Are KU books more likely to be in the low-price / mid-price segments (0–9.99, 10–14.99) compared to high price?

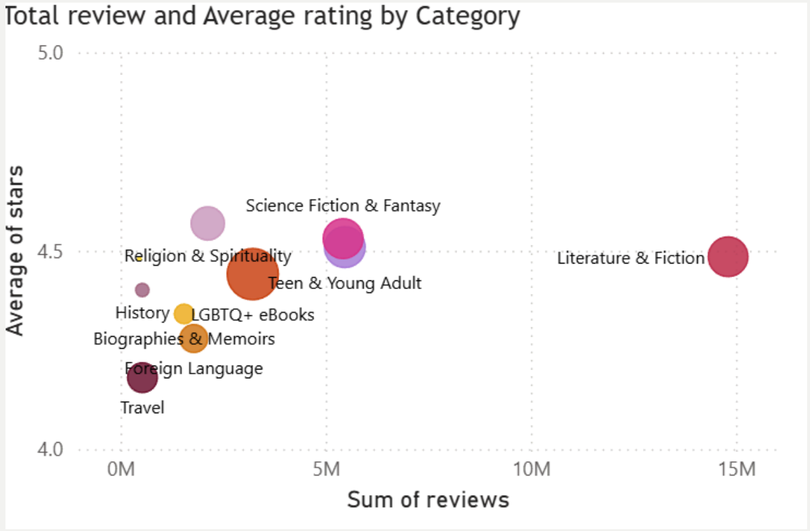
TOTAL PRODUCT

72.61K	118M	131.84K	4.40
Total Authors	Total Reviews	Titles	Average Rating

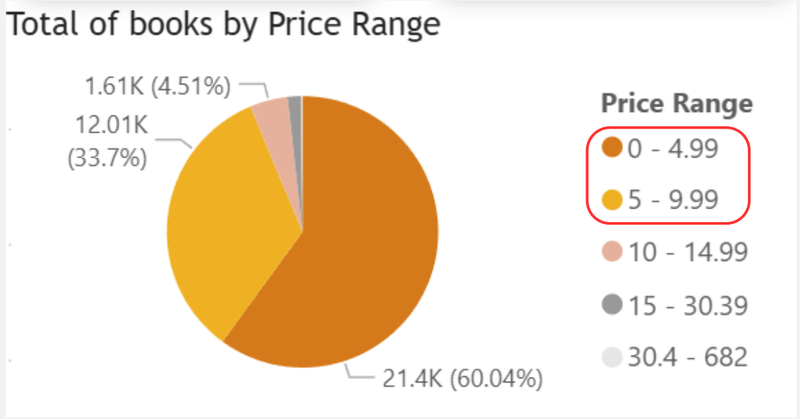




KINDLE UNLIMITED PERFORMANCE



72.61K	118M	131.84K	4.40
Total Authors	Total Reviews	Titles	Average Rating



- **KU ≈ 27% of Total of Books** and ≈ **32% of Reviews;** Avg Rating ≈ 4.4★ (same as total).
- **~94% of KU Books < $10**
- KU has high volume and interaction in **key genres** such as **Literature & Fiction**

KU succeeds when affordable pricing meets bingeable categories

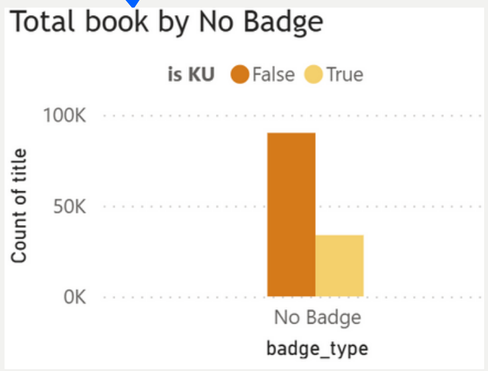
DATA ANALYSIS

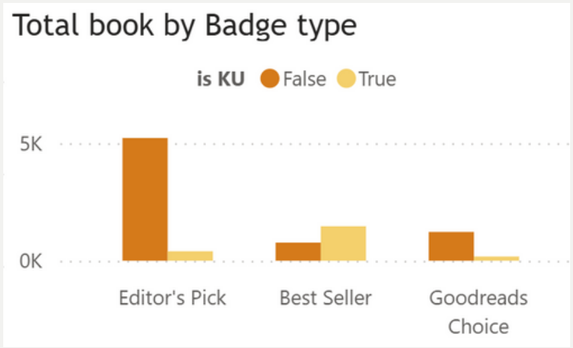
Product

- Do KU book rely more heavily on certain **badges** than non-KU books?

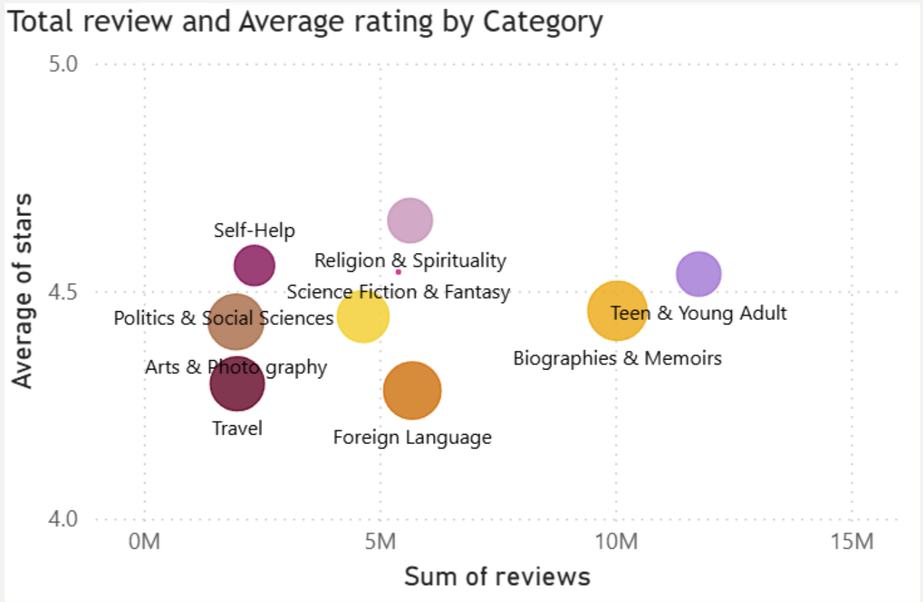
TOTAL PRODUCT

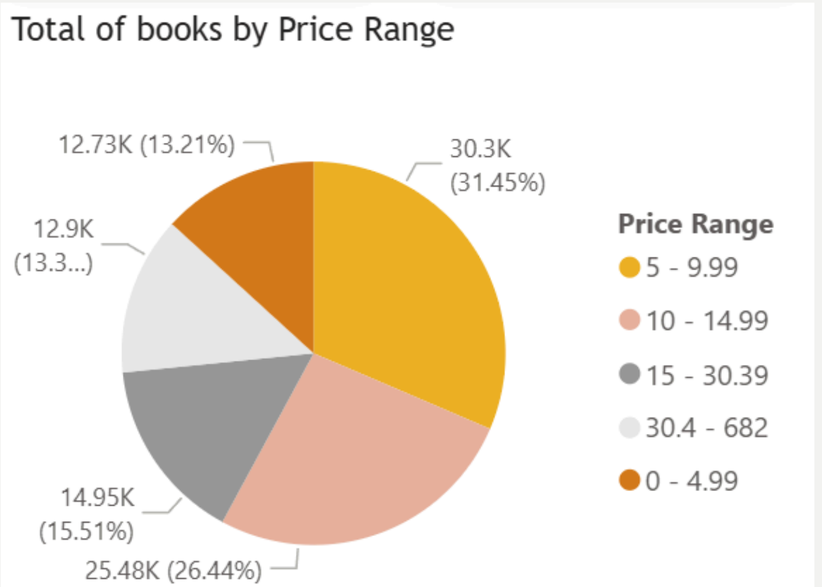






NON-KU





- **High Review** and **high Rating** such as Biographies & Memoirs and Teen & Young Adult
- Non-KU price pie shows a heavy **high-price mix**

KU growth by converting **top titles** from these **non KU,** well-rated, top reviews and expand pricing **$10-14.99**

DATA ANALYSIS

User Engagement

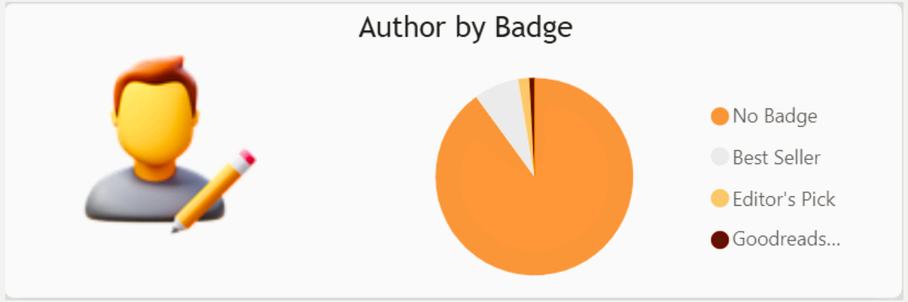
IN KINDLE UNLIMITED

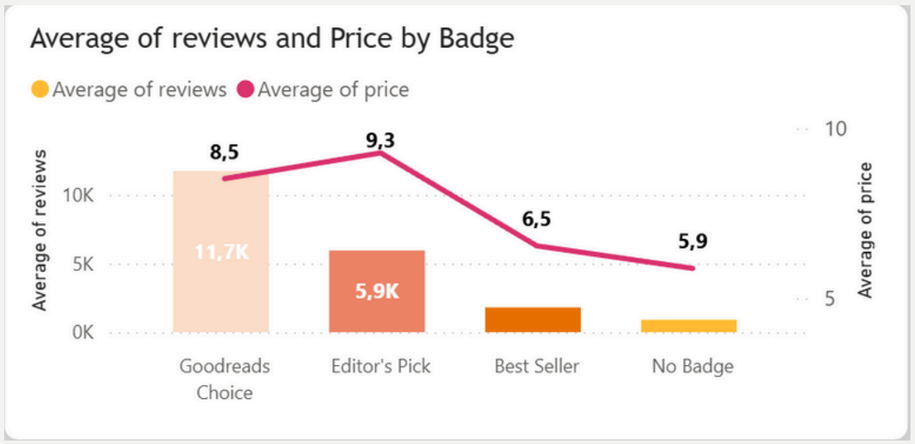
72,61K	22%	38M	35,75K	4,40
Total Authors	% Authors by KU	Total Reviews	Titles	Average of stars

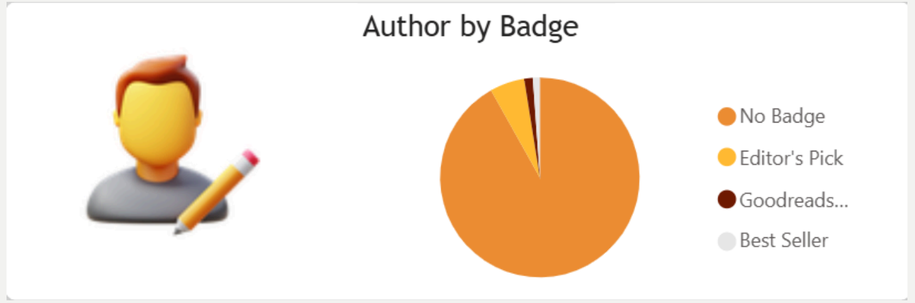
NON-KINDLE UNLIMITED

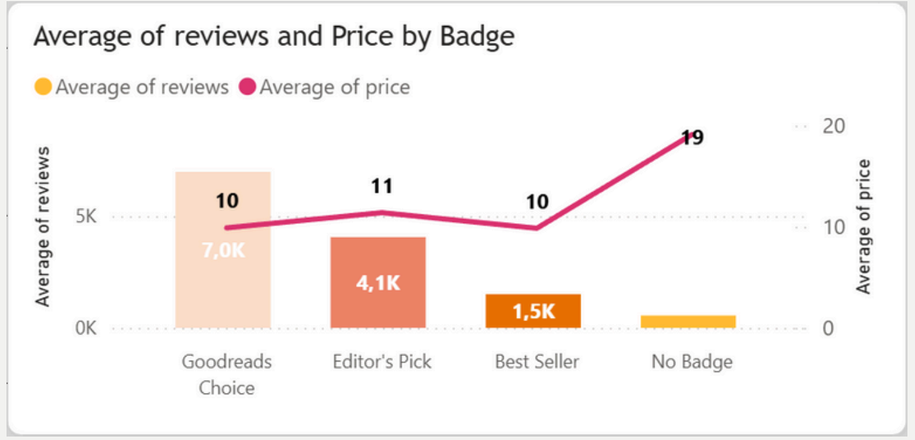
72,61K	81%	81M	96,16K	4,41
Total Authors	% Authors by KU	Total Reviews	Titles	Average of stars

Non-KU is much larger, with more authors, more books, and nearly double the reviews







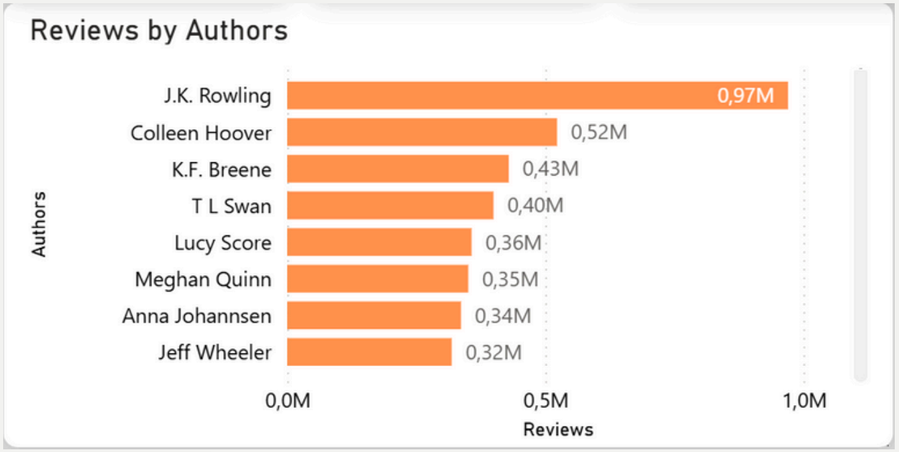


KU, despite being smaller, delivers higher reviews per title, especially for Goodreads Choice and Editor's Pick Badge authors.

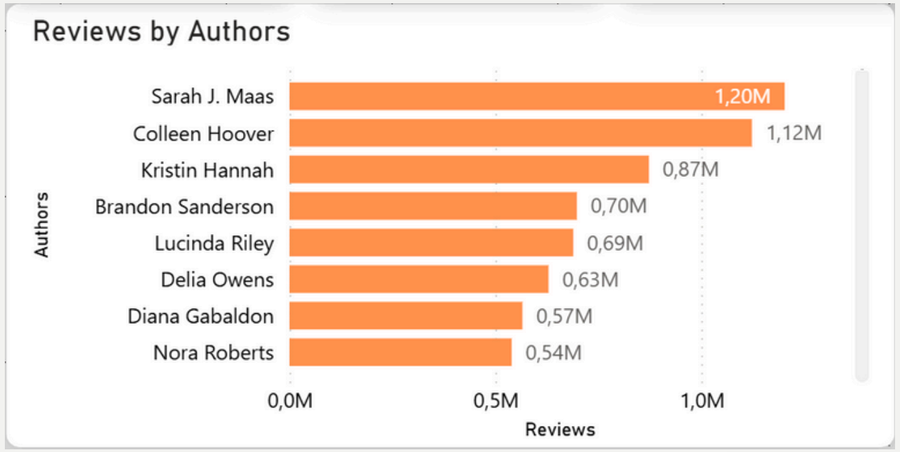
DATA ANALYSIS

User Engagement

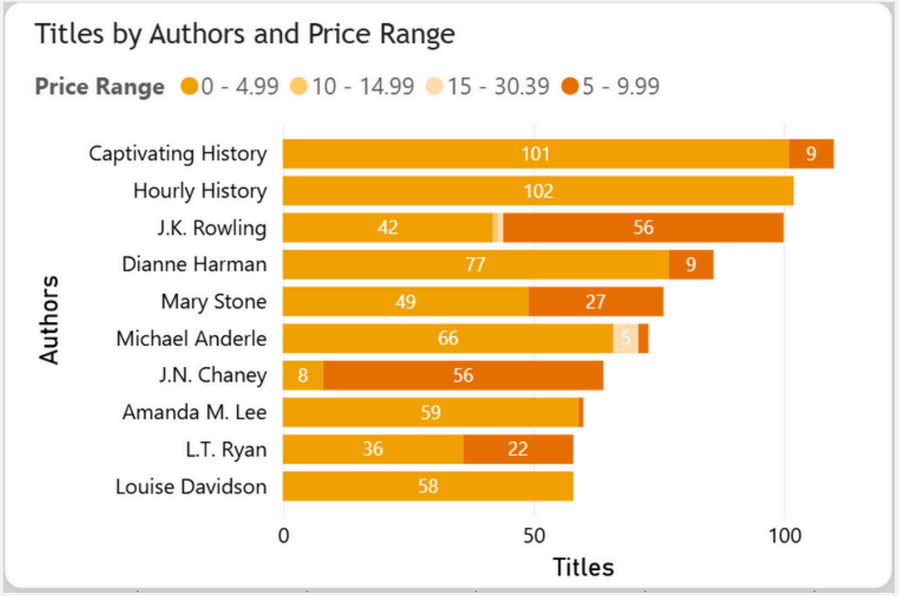
IN KINDLE UNLIMITED

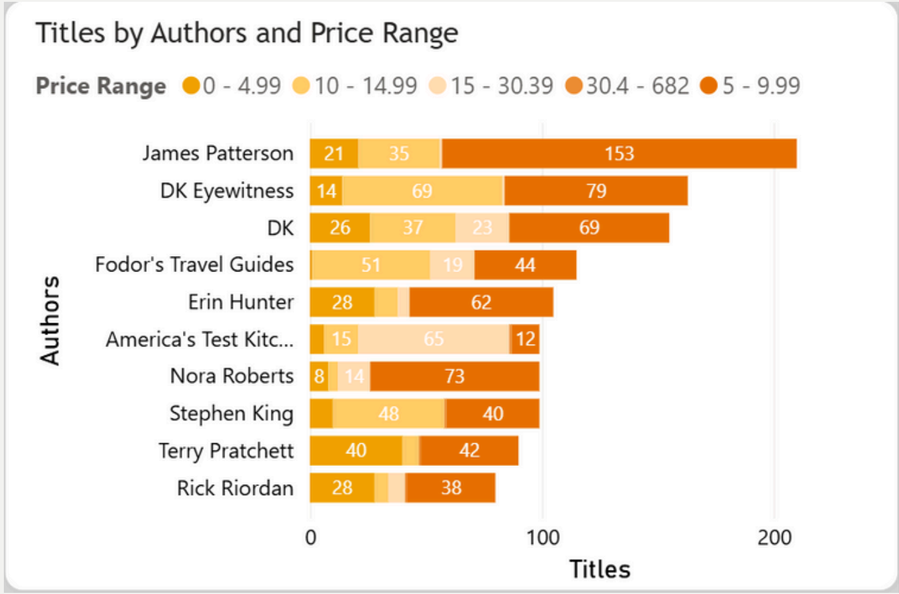


NON-KINDLE UNLIMITED



KU is driven by a few superstar authors, while Non-KU has a broader spread across many big names.





Pricing: Non-KU has more premium-priced titles, averaging 18 to 24 dollars, while KU titles are consistently low-priced at 5 to 10 dollars.

DATA ANALYSIS

User Engagement

- Scale: Non-KU is much larger, with more authors, more titles, and nearly double the reviews.
- Efficiency: KU, despite its smaller size, delivers higher reviews per title—especially for authors without Goodreads Choice and Editor's Pick Badge.
- Pricing: KU titles are consistently low-priced at under $10, while Non-KU titles range higher, averaging $10–$30.
- Author Dynamics: KU depends on a few superstar authors, while Non-KU distributes influence across a broader set of big names.

Strategic Insight:

- Potential expansion across all badges with good reviews
- Authors having positive interactions from users can boost visibility and engagement for Kindle Unlimited

Top 10 Authors with Editor's Pick Badge

Authors	Reviews	Rating
Brené Brown	40680	4,70
Cassandra Clare	24992	4,60
Alice Oseman	24412	4,70
Raina Telgemeier	24056	4,70
Ben Aaronovitch	21159	4,30
F. Scott Fitzgerald	17003	4,40
Jason Reynolds	16750	4,75
Naomi Novik	14982	4,60
Barbara Kingsolver	14593	4,50
Louise L. Hay	14487	4,70
Total	**213114**	**4,61**

Top 10 Authors with Goodreads Choice Badge

Authors	Reviews	Rating
Yuval Noah Harari	135224	4,60
Trevor Noah	103633	4,70
E L James	75976	4,50
Cheryl Strayed	72700	4,40
Veronica Roth	57685	4,50
Susanna Clarke	25942	4,50
Stefania Auci	24594	4,40
Jojo Moyes	24153	4,45
Cassandra Clare	23310	4,70
Robert Harris	21952	4,20
Total	**565169**	**4,51**

Top 10 Authors with No Badge

Authors	Reviews	Rating
Terry Pratchett	25699	4,72
Shelby Mahurin	14315	4,40
Marianne Cronin	12988	4,50
Maggie O'Farrell	21790	4,20
Kim Michele Richardson	50356	4,60
Karen M. McManus	18158	4,60
Douglas Adams	23993	4,40
Chris Kyle	42409	4,50
Cassandra Clare	19670	4,70
Austin Kleon	16873	4,60
Total	**246251**	**4,59**

Top 10 Authors with No Badge

Authors	Reviews	Rating
Sarah J. Maas	1201387	4,70
Colleen Hoover	1119948	4,54
Kristin Hannah	871214	4,48
Brandon Sanderson	692826	4,66
Lucinda Riley	676493	4,49
Delia Owens	628949	4,68
Diana Gabaldon	565624	4,64
Nora Roberts	539621	4,55
Taylor Jenkins Reid	516495	4,37
Julia Quinn	448383	4,51
Total	**7260940**	**4,56**

PROPOSED SOLUTIONS

Key insights



Risk of Author Concentration

Over-reliance on a small group creates risk. Need to leverage their influence while nurturing new talent



Untapped Premium Segment

KU is missing a premium audience. A carefully selected higher-priced segment could attract new users without hurting value perception

Power of Badges as Quality Signals

Badges are a powerful signal of quality and drive discovery. KU should maximize its use



Winning Genre Strategy



KU succeeds where affordable pricing meets high-engagement genres. Content acquisition should focus on these areas

PROPOSED SOLUTIONS

Strategy for Kindle Unlimited

01

Premier Author Program

- **Focus**:
  Secure exclusives with top authors & develop new talent.
- **Goal**:
  Ensure a steady pipeline of must-have content to drive loyalty.

02

Premium Catalog Pilot

- **Focus**:
  Add select high-quality books in the $10-14.99 range.
- **Goal**:
  Attract premium subscribers and increase revenue per user.

03

"KU Gem" Badge

- **Focus**:
  Algorithmically tag high-rated books without official badges.
- **Goal**:
  Boost discovery of quality content and increase user engagement.

04

Category Domination

- **Focus**:
  Acquire top-rated titles in key genres (Literature Fiction, Teen & Young Adult).
- **Goal**:
  Become the leading platform for bingeable reading and promote growth.

PROPOSED SOLUTIONS

Implementation

PREMIER AUTHOR PROGRAM



- Identify top-tier authors (≥4.3★, high reviews). Offer exclusive contracts for new series.
- Create a "Rising Star" program with support for promising new authors.
- Launch marketing campaigns featuring these exclusive authors.

PREMIUM CATALOG PILOT



- Select top high-rated books in Business/Finance priced at $10-14.99.
- Target marketing to high-income users.
- Track subscriber growth and engagement from this segment.

"KU GEM" BADGE

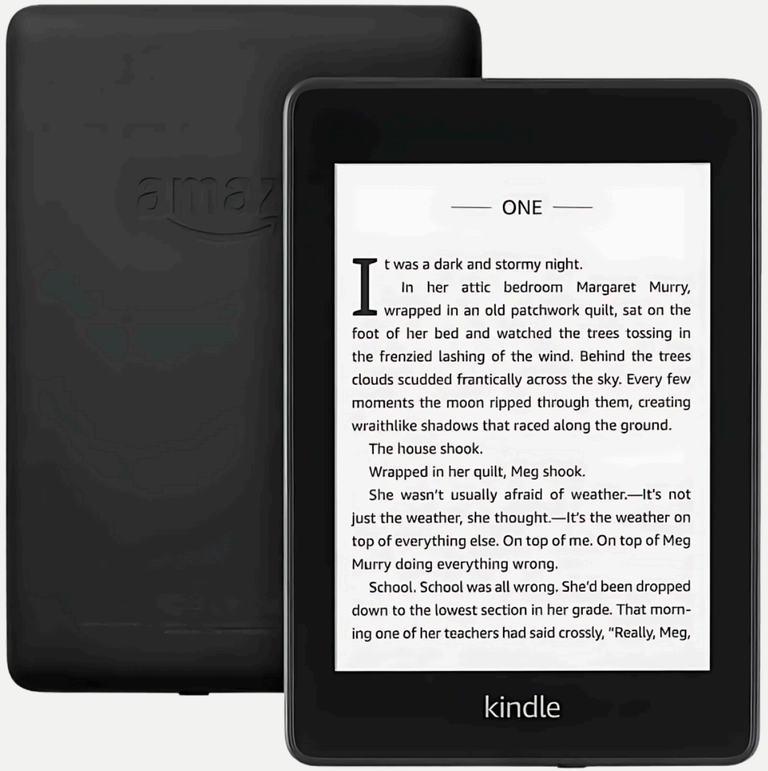


- Develop algorithm to auto-tag high-quality books without badges.
- Display badge on book covers and create "Hidden Gems" section.
- Feature these books in recommendations and newsletters.

CATEGORY DOMINATION



- Identify top-rated Non-KU books in Literature Fiction, Teen & Young Adult
- Acquire exclusive KU rights for these titles.
- Promote through themed campaigns like "Bestsellers Now on KU".



Thank you

Knock Knock

Who's there?

Column.

Column who?

Column me mistaken but I love data!